

PALESTINE, TX, USA

WMO: 720314

Lat: 31.780N

Lon: 95.706W

Elev: 129

StdP: 99.79

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 93983

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	-1.3	-10.9	1.5	3.2	-8.0	1.9	4.6	9.4	11.1	8.4	12.5	1.7	340	0.389

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.6	37.6	23.1	36.4	23.3	35.2	23.5	26.2	31.7	25.8	31.1	25.4	30.7	2.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	20.5	27.9	24.6	19.9	27.7	24.1	19.3	27.4	82.0	31.8	80.3	31.3	78.6	30.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.9	7.0	5.8	-7.4	39.3	2.5	1.5	-9.2	40.4	-10.7	41.3	-12.1	42.1	-13.9	43.2		
			-8.0	26.7	2.2	0.7	-9.6	27.2	-10.9	27.6	-12.2	27.9	-13.8	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.7	9.3	11.5	15.6	19.0	23.2	27.2	28.7	29.2	26.3	20.6	14.6	10.4
	DBStd	8.23	5.52	5.94	5.28	4.33	3.51	2.21	1.85	2.12	2.83	4.56	5.36	5.46
	HDD10.0	228	79	49	15	2	0	0	0	0	0	1	19	63
	HDD18.3	1068	284	201	114	44	7	0	0	0	1	31	134	252
	CDD10.0	3757	58	91	189	271	409	516	579	596	488	328	156	76
	CDD18.3	1556	4	9	31	63	157	266	321	338	238	100	22	7
	CDH23.3	16473	18	46	176	486	1412	2912	3769	4194	2450	871	114	24
	CDH26.7	7748	1	3	23	106	506	1403	1948	2291	1160	292	15	1
Wind	WSAvg	2.5	2.8	3.1	3.2	3.1	2.7	2.3	2.1	2.0	2.0	2.3	2.5	2.6
Precipitation	PrecAvg	1187	103	101	97	94	98	109	68	85	94	114	97	107
	PrecMax	1662	208	229	181	221	199	252	275	245	194	315	319	232
	PrecMin	676	9	3	24	19	0	0	0	37	0	5	14	
	PrecStd	270	55	59	42	51	60	69	59	68	46	86	69	61
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.3	27.2	28.9	32.2	34.1	37.6	38.5	40.3	37.7	33.0	28.5	26.0
		MCWB	17.0	17.2	18.7	21.8	23.0	22.8	22.8	22.9	21.3	22.2	21.1	18.6
	2%	DB	22.9	24.9	27.2	29.3	32.6	35.8	37.2	37.8	35.8	31.5	26.3	23.5
		MCWB	16.5	17.3	18.8	20.2	22.7	22.9	23.0	23.6	22.5	22.0	19.5	18.3
	5%	DB	21.2	22.6	25.3	27.6	31.1	34.2	35.9	36.9	33.9	30.0	24.5	21.6
		MCWB	16.5	17.1	18.1	19.7	22.6	23.5	23.3	23.8	23.0	21.5	18.8	17.7
	10%	DB	18.9	20.9	23.6	26.2	29.8	32.9	34.5	35.5	32.5	28.0	22.7	20.0
		MCWB	14.9	16.5	17.1	19.2	22.1	23.8	24.1	23.9	22.8	20.5	18.6	16.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.9	20.7	21.4	23.7	25.3	26.7	26.7	26.8	26.1	25.3	22.7	21.4
		MCDB	22.8	23.4	25.3	27.6	29.7	31.8	31.9	31.8	31.2	28.9	25.7	23.1
	2%	WB	18.8	19.7	20.5	22.4	24.4	25.9	26.2	26.2	25.5	24.3	21.6	19.8
		MCDB	21.7	22.2	24.4	26.9	29.2	30.9	31.5	31.9	30.4	28.3	24.4	22.3
	5%	WB	17.6	18.5	19.6	21.6	23.8	25.4	25.7	25.7	25.0	23.6	20.6	18.7
		MCDB	20.0	21.5	23.7	25.8	28.6	30.3	31.0	31.4	29.8	27.3	23.2	21.4
	10%	WB	15.3	17.1	18.8	20.7	23.2	24.9	25.3	25.3	24.5	22.8	19.3	17.2
		MCDB	18.0	20.2	22.4	24.5	27.9	29.6	30.5	30.9	29.0	26.1	22.2	19.4
Mean Daily Temperature Range	5% DB	MDBR	11.4	10.9	11.4	12.0	11.1	10.6	11.1	11.6	11.4	12.1	11.6	10.9
		MCDBR	13.1	12.0	11.6	12.2	12.0	12.2	13.1	13.5	13.0	12.4	11.7	12.0
		MCWBR	7.6	7.1	5.7	5.5	4.3	3.2	2.8	3.0	3.7	4.7	6.4	8.0
	5% WB	MCDBR	10.5	10.1	10.0	10.1	10.2	9.9	10.2	11.1	10.3	9.9	10.0	10.2
	MCWBR	8.0	7.0	6.1	5.5	4.5	3.5	3.1	3.1	3.5	4.6	6.7	8.1	
Clear-Sky Solar Irradiance	taub	0.320	0.341	0.363	0.405	0.440	0.449	0.456	0.458	0.416	0.364	0.353	0.327	
	taud	2.512	2.464	2.408	2.303	2.258	2.275	2.289	2.275	2.346	2.471	2.451	2.500	
	Ebn at Noon	907	918	918	887	854	842	834	828	853	880	857	877	
	Edh at Noon	87	100	113	131	138	135	133	133	119	98	92	84	
All-Sky Solar Radiation	RadAvg	2.81	3.17	4.34	5.37	5.79	6.37	6.48	6.07	5.12	4.14	3.02	2.46	
	RadStd	0.26	0.56	0.40	0.36	0.43	0.54	0.45	0.43	0.45	0.58	0.38	0.32	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
										+113	+88

Nomenclature: See separate page